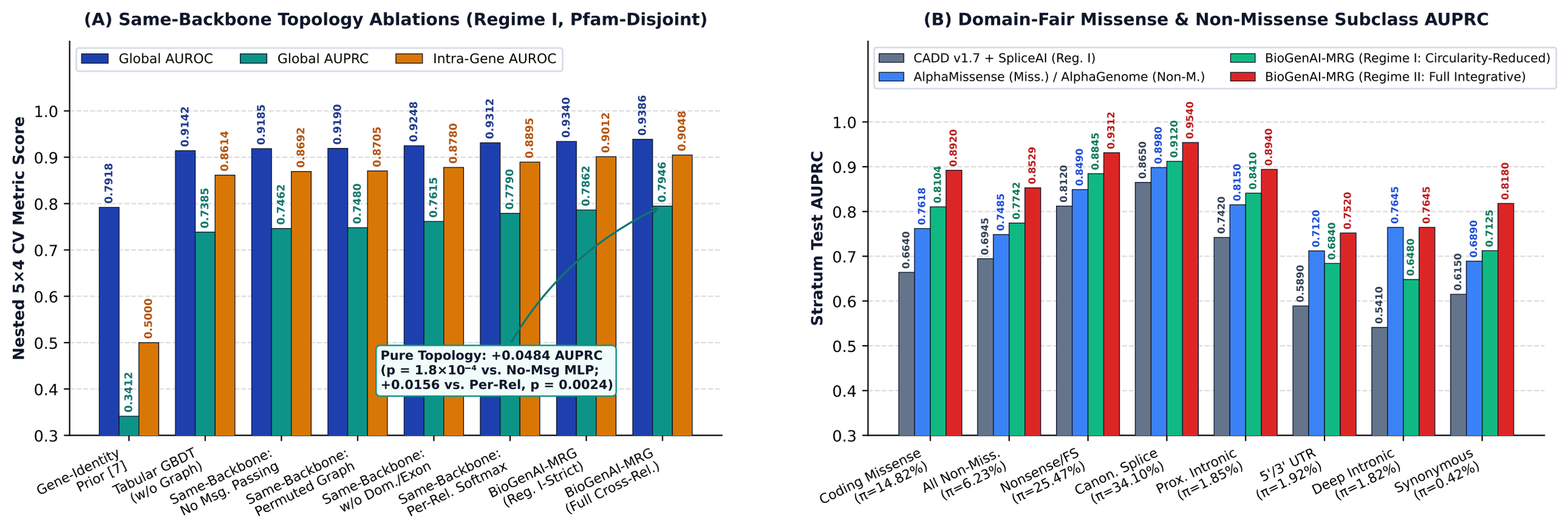
(A) Same-Backbone Topology Ablations (Regime I, Pfam-Disjoint)
Global AUROC
Global AUPRC
Intra-Gene AUROC
Nested 5×4 CV Metric Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.7918
0.3412
0.5000
0.9142
0.7385
0.8614
0.9185
0.7462
0.8692
0.9190
0.7480
0.8705
0.9248
0.7615
0.8780
0.9312
0.7790
0.8895
0.9340
0.7862
0.9012
0.9386
0.7946
0.9048
Pure Topology: +0.0484 AUPRC
(p = 1.8×10⁻⁴ vs. No-Msg MLP;
+0.0156 vs. Per-Rel, p = 0.0024)
Gene-Identity Prior [7]
Tabular GBDT (w/o Graph)
Same-Backbone: No Msg. Passing
Same-Backbone: Permuted Graph
Same-Backbone: w/o Dom./Exon
Same-Backbone: Per-Rel. Softmax
BioGenAI-MRG (Reg. I-Strict)
BioGenAI-MRG (Full Cross-Rel.)
(B) Domain-Fair Missense & Non-Missense Subclass AUPRC
CADD v1.7 + SpliceAI (Reg. I)
AlphaMissense (Miss.) / AlphaGenome (Non-M.)
BioGenAI-MRG (Regime I: Circularity-Reduced)
BioGenAI-MRG (Regime II: Full Integrative)
Stratum Test AUPRC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.6640
0.7618
0.8104
0.8920
0.6945
0.7485
0.7742
0.8529
0.8120
0.8490
0.8845
0.9312
0.8650
0.8980
0.9120
0.9540
0.7420
0.8150
0.8410
0.8940
0.5890
0.7120
0.6840
0.7520
0.5410
0.7645
0.6480
0.7645
0.6150
0.6890
0.7125
0.8180
Coding Missense (π=14.82%)
All Non-Miss. (π=6.23%)
Nonsense/FS (π=25.47%)
Canon. Splice (π=34.10%)
Prox. Intronic (π=1.85%)
5′/3′ UTR (π=1.92%)
Deep Intronic (π=1.82%)
Synonymous (π=0.42%)